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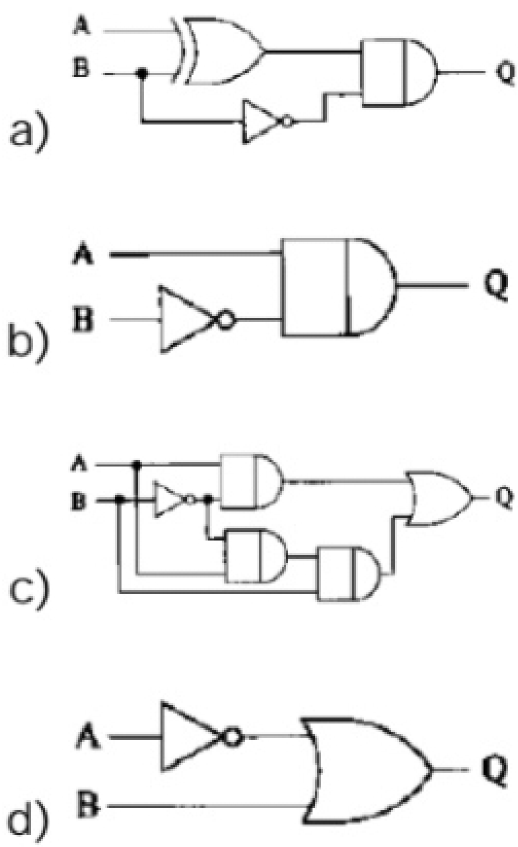
GATE Question No. 41

Question:
For any set of inputs, A and B, the following circuits give the same output, Q, except one. Identify which one is different.



(GATE PH 2010)

Solution

We simplify each circuit using Boolean algebra.

Step 1: Circuit (a)
From the diagram, output is AND of $(A + B)$ and B'.
$Q_a = (A + B) \cdot B'$
Using distributive law:
$Q_a = AB' + BB'$
Since $BB' = 0$,
$Q_a = AB'$

Step 2: Circuit (b)
$Q_b = AB'$

Step 3: Circuit (c)
After simplifying internal gates,
$Q_c = AB'$

Step 4: Circuit (d)
$Q_d = A' + B$

Truth Table:

A	B	AB'	$A' + B$
0	0	0	1
0	1	0	1
1	0	1	0
1	1	0	1

Circuits (a), (b), and (c) give same output.
Circuit (d) gives different output.

Correct Answer: Option (d)